

```

1 //@version=5
2 // =====
3 // AURUM EDGE V5 - XAUUSD 30m - FINAL
4 //
5 // The consolidated bot. Supersedes V3 and V4: a rebuilt long engine (simpler AND
6 // better than V3's), plus V4's two short engines kept intact but off by default.
7 //
8 // WHAT CHANGED FROM V3, AND WHY
9 //   V3 waited for a confirming close back above the EMA before buying a dip, so
10 //   it entered after the bounce had already begun. On an asset with structural
11 //   upward drift that confirmation costs more than it saves. Dropping it - just
12 //   buying into two consecutive red bars while the trend stack is up - was
13 //   better on every measure, and it needs only two entry rules instead of three.
14 //
15 //   Walk-forward, 16 folds, out-of-sample only, exit re-chosen each fold:
16 //       V3 (3 entries)           2.40 trades/day E[R] +0.284 +572 R maxDD 47.5
17 //       V5 (2 entries)           2.75 trades/day E[R] +0.290 +668 R maxDD 43.0
18 //   More trades, higher expectancy, ~100 R more, LESS drawdown, fewer rules.
19 //   Return/drawdown improves from 12.0 to 15.5.
20 //
21 // -----
22 // READ THIS BEFORE YOU ENABLE THE SHORT ENGINES
23 //
24 // Both short engines are OFF by default, and that is a measurement, not caution.
25 // They were built to spec and tested on 1,678,872 one-minute bars (Jan 2021 -
26 // Aug 2026) with real historical spreads, and the daily bear gate was ACTIVE FOR
27 // 18.6% of that history (10,411 of 55,988 30m bars, longest run 123 days). So
28 // this was a fair test on a real sample, not an untested regime.
29 //
30 //   SYMMETRIC SHORT ENGINE (daily bear gate, 3 mirrored triggers)
31 //       mirrored ladder exit  n=491 E[R] -0.086 PF 0.85 t=-1.28
32 //       BE@1R + 2R target     n=638 E[R] -0.007 PF 0.99 t=-0.14 <- least bad
33 //       fixed 1R target       n=736 E[R] -0.024 PF 0.95 t=-0.65
34 //       4xATR trail           n=573 E[R] -0.080 PF 0.84 t=-1.56
35 //       on 1h bars            n=321 E[R] -0.136 PF 0.78
36 //       on 4h bars            n=117 E[R] -0.215 PF 0.68
37 //
38 //   COUNTER-TREND SHORT (overbought spike inside a bull regime)
39 //       ALL 18 variants tested were negative, most significantly so:
40 //       ext>1.5ATR RSI>70 tp1R  n=575 E[R] -0.206 t=-5.19 p=1.000
41 //       ext>2.0ATR RSI>75 ladder n=225 E[R] -0.399 t=-5.31 p=1.000
42 //
43 // WHY. Gold is not symmetric. It carries structural upward drift, so its
44 // downtrends are shallower, choppy and revert faster than its uptrends. A
45 // mirrored trend-follower assumes downmoves persist the way upmoves do; on gold
46 // they do not. The pullbacks that look like short setups on a daily chart are
47 // mostly mean-reverting dips inside an uptrend - which is precisely what the
48 // LONG engine is buying, and why shorting them loses.
49 //
50 // The engines are here, complete and symmetric, so you can forward-test them
51 // yourself and so the machinery exists if gold ever breaks its multi-year
52 // uptrend. At that point the short side is worth re-testing on new data. Until
53 // then, the evidence says leave them off.
54 //
55 // -----
56 // LONG ENGINE - walk-forward validated, out-of-sample only
57 //   16 folds, Jan 2023 - Mar 2026, exit re-chosen inside each training window
58 //       2.75 trades/day n=2307 E[R] +0.290 R total +668.2 R
59 //       win 35.5% PF 1.50 maxDD 43.0 R t=+7.52 p<0.0001
60 //       deflated Sharpe P(SR>0) = 1.000
61 //   At 0.25% risk: ~+48%/yr at ~13% drawdown, against buy-and-hold gold at
62 //   +47%/yr and 19% drawdown over the same window.
63 //
64 //   Caveat, stated plainly: all 2,011 trades are long and the out-of-sample
65 //   window is a gold bull market. The least-trending slice (2023) returned
66 //   +0.118 R at p=0.066, which is not significant.
67 // =====
68
69 strategy(
70     "Aurum Edge V5 - XAUUSD 30m Final",
71     overlay           = true,
72     pyramiding         = 10,
73     initial_capital     = 10000,
74     currency           = currency.USD,
75     default_qty_type    = strategy.percent_of_equity,
76     default_qty_value   = 100,
77     process_orders_on_close = false,
78     calc_on_every_tick  = false,
79     calc_on_order_fills = false,
80     commission_type     = strategy.commission.cash_per_contract,
81     commission_value    = 0.02,
82     slippage            = 3,
83     max_labels_count    = 500)
84
85 // =====
86 // INPUTS
87 // =====
88 string G_ENG = "1. Engines"
89 string G_R   = "2. Risk"

```

```

90 string G_LE = "3. Long entries (bull regime)"
91 string G_SE = "4. Short entries (bear regime)"
92 string G_CT = "5. Counter-trend shorts (bull overbought)"
93 string G_LAD = "6. Ladder exit"
94 string G_UI = "7. Display and alerts"
95
96 enableLong = input.bool(true, "Enable Long Engine (bull regime)", group = G_ENG)
97 enableShort = input.bool(false, "Enable Symmetric Short Engine (bear regime)", group = G_ENG,
98     tooltip = "OFF by default because it was measured at -0.086 R per trade over 491 " +
99         "trades in a genuine bear sample (18.6% of history). Enable to forward-test.")
100 enableCT = input.bool(false, "Enable Counter-Trend Short Engine (bull overbought)", group = G_ENG,
101     tooltip = "OFF by default. All 18 tested variants were negative, most at t < -4. " +
102         "Shorting overbought gold inside an uptrend lost money reliably.")
103
104 riskPct = input.float(0.25, "Risk per trade (% of equity)", 0.05, 3.0, 0.05, group = G_R)
105 ctRiskMult = input.float(0.5, "Counter-trend risk multiplier", 0.1, 1.0, 0.1, group = G_R,
106     tooltip = "Counter-trend trades size smaller than trend trades.")
107 stopAtr = input.float(2.0, "Initial stop (ATR)", 0.5, 6.0, 0.1, group = G_R)
108 minStopUsd = input.float(8.0, "Minimum stop ($)", 1.0, 60.0, 0.5, group = G_R,
109     tooltip = "Cost drag = round-trip cost / stop distance. Round trip on gold is " +
110         "roughly $0.45-$1.20; below about $5 of stop the spread eats the edge.")
111 maxStopUsd = input.float(200.0, "Maximum stop ($)", 20.0, 500.0, 5.0, group = G_R)
112 maxOpen = input.int(5, "Max concurrent positions (per side)", 1, 10, group = G_R)
113
114 useL1 = input.bool(true, "L1: unconfirmed dip (N red bars in an uptrend)", group = G_LE,
115     tooltip = "The V5 change. Buys INTO the dip rather than waiting for a close back " +
116         "above the EMA. Tested +0.365 R over 1,587 trades against +0.351 R over " +
117         "1,005 for V3's confirmed version: better expectancy AND 58% more trades.")
118 dipBars = input.int(2, "red bars required", 1, 5, group = G_LE)
119 useL3 = input.bool(true, "L2: 24-bar breakout", group = G_LE)
120 useLegacyPb = input.bool(false, "L3: V3 legacy confirmed pullback (optional)", group = G_LE,
121     tooltip = "V3's original entry. Adding it back gives 2.81 trades/day at +0.297 R but " +
122         "raises drawdown from 43.0 to 48.3 R. Off by default: two rules beat three.")
123
124 useS1 = input.bool(true, "S1: rally pullback to EMA20", group = G_SE)
125 useS2 = input.bool(true, "S2: deep rally pullback to EMA50", group = G_SE)
126 useS3 = input.bool(true, "S3: 24-bar breakdown", group = G_SE)
127 shortExitMode = input.string("Tight 1R target", "Short exit style", group = G_SE,
128     options = ["Mirrored ladder", "Tight 1R target", "BE at 1R then 2R target"],
129     tooltip = "Trend-following exits were the WORST on shorts (-0.086 R) while a tight " +
130         "target was least bad (-0.007 R). The asymmetry is the point: gold's " +
131         "downmoves do not persist the way its upmoves do.")
132
133 ctExtAtr = input.float(2.0, "Extension above EMA20 (ATR)", 0.5, 6.0, 0.25, group = G_CT)
134 ctRsi = input.float(70, "RSI threshold", 50, 95, 1, group = G_CT)
135 ctTpr = input.float(1.0, "Counter-trend target (R)", 0.5, 5.0, 0.25, group = G_CT)
136
137 r1Trig = input.float(1.5, "Rung 1 trigger (R)", 0.25, 5.0, 0.25, group = G_LAD)
138 r1Lock = input.float(0.0, "Rung 1 lock (R)", -1.0, 5.0, 0.1, group = G_LAD, tooltip = "0 = breakeven")
139 r2Trig = input.float(3.0, "Rung 2 trigger (R)", 0.5, 9.0, 0.25, group = G_LAD)
140 r2Lock = input.float(1.5, "Rung 2 lock (R)", -1.0, 9.0, 0.1, group = G_LAD)
141 r3Trig = input.float(5.0, "Rung 3 trigger (R)", 1.0, 15.0, 0.25, group = G_LAD)
142 r3Lock = input.float(3.2, "Rung 3 lock (R)", -1.0, 15.0, 0.1, group = G_LAD)
143 maxHoldBars = input.int(96, "Time stop (bars)", 8, 500, group = G_LAD, tooltip = "96 x 30m = 48h")
144
145 atrLen = input.int(14, "ATR length", 2, 100, group = G_UI)
146 showLabels = input.bool(true, "Show trade labels", group = G_UI)
147 showPanel = input.bool(true, "Show regime panel", group = G_UI)
148 tintBg = input.bool(true, "Tint background by regime", group = G_UI)
149
150 // =====
151 // HIGHER-TIMEFRAME REGIME GATES
152 // Only COMPLETED daily bars are read ([1] offset), so these cannot repaint.
153 // =====
154 dE20 = request.security(syminfo.tickerid, "D", ta.ema(close, 20)[1], lookahead = barmerge.lookahead_on)
155 dE50 = request.security(syminfo.tickerid, "D", ta.ema(close, 50)[1], lookahead = barmerge.lookahead_on)
156 dE200 = request.security(syminfo.tickerid, "D", ta.ema(close, 200)[1], lookahead = barmerge.lookahead_on)
157
158 bool macroBull = dE20 > dE50 and dE50 > dE200 // long gate
159 bool macroBear = dE20 < dE50 and dE50 < dE200 // exact inverse
160
161 // =====
162 // 30-MINUTE ENGINE
163 // =====
164 ema20 = ta.ema(close, 20)
165 ema50 = ta.ema(close, 50)
166 ema200 = ta.ema(close, 200)
167 atr = ta.atr(atrLen)
168 rsi = ta.rsi(close, 14)
169
170 plot(ema20, "EMA20", color = color.new(color.aqua, 0), linewidth = 1)
171 plot(ema50, "EMA50", color = color.new(color.orange, 0), linewidth = 1)
172 plot(ema200, "EMA200", color = color.new(color.purple, 0), linewidth = 2)
173
174 bool stackUp = ema20 > ema50 and ema50 > ema200
175 bool stackDn = ema20 < ema50 and ema50 < ema200
176 bool slopeUp = ema20 > ema20[3]
177 bool slopeDn = ema20 < ema20[3]
178

```

```

179 float hh24 = ta.highest(high, 24)[1]
180 float ll24 = ta.lowest(low, 24)[1]
181
182 // ---- LONG triggers -----
183 // L1 is the V5 change: buy INTO the dip, no confirming close required.
184 int redRun = 0
185 for i = 0 to dipBars - 1
186     redRun += close[i] < open[i] ? 1 : 0
187 bool sigL1 = useL1 and stackUp and redRun == dipBars
188 bool sigL3 = useL3 and close > hh24 and close > open
189 bool sigL2 = useLegacyPb and stackUp and slopeUp and low <= ema20 and close > ema20 and close > open
190 bool longSignal = enableLong and macroBull and (sigL1 or sigL2 or sigL3)
191
192 // ---- SHORT triggers (exact structural mirrors) -----
193 bool sigS1 = useS1 and stackDn and slopeDn and high >= ema20 and close < ema20 and close < open
194 bool sigS2 = useS2 and stackDn and high >= ema50 and close < ema50 and close < open
195 bool sigS3 = useS3 and close < ll24 and close < open
196 bool shortSignal = enableShort and macroBear and (sigS1 or sigS2 or sigS3)
197
198 // ---- COUNTER-TREND short (overbought exhaustion inside a bull regime) -----
199 float extAtr = atr > 0 ? (close - ema20) / atr : 0.0
200 bool sigCT = enableCT and macroBull and extAtr >= ctExtAtr and rsi >= ctRsi and close < open
201
202 // =====
203 // POSITION SIZING
204 // =====
205 float stopDist = math.min(math.max(atr * stopAtr, minStopUsd), maxStopUsd)
206 float qtyBase = stopDist > 0 ? (strategy.equity * riskPct / 100.0) / stopDist : na
207 float qtyCT = na(qtyBase) ? na : qtyBase * ctRiskMult
208
209 int openLongs = 0
210 int openShorts = 0
211 if strategy.opentrades > 0
212     for i = 0 to strategy.opentrades - 1
213         if strategy.opentrades.size(i) > 0
214             openLongs += 1
215         else
216             openShorts += 1
217
218 // =====
219 // ENTRIES - each position carries its own id so it can run its own ladder
220 // =====
221 var int seq = 0
222 var map<string, float> riskOf = map.new<string, float>()
223 var map<string, float> entryOf = map.new<string, float>()
224 var map<string, int> barOf = map.new<string, int>()
225 var map<string, int> kindOf = map.new<string, int>() // 0 long, 1 short, 2 counter-trend
226
227 _register(string id, float rp, float ep, int kind) =>
228     map.put(riskOf, id, rp)
229     map.put(entryOf, id, ep)
230     map.put(barOf, id, bar_index)
231     map.put(kindOf, id, kind)
232
233 if longSignal and openLongs < maxOpen and not na(qtyBase) and qtyBase > 0
234     seq += 1
235     string id = "L" + str.toString(seq)
236     strategy.entry(id, strategy.long, qty = qtyBase)
237     _register(id, stopDist, close, 0)
238     if showLabels
239         label.new(bar_index, low, "L", style = label.style_label_up,
240             color = color.new(color.green, 20), textcolor = color.white, size = size.tiny)
241     alert("AURUM_V5 LONG entry=" + str.toString(close, "#.##") +
242         " stop=" + str.toString(close - stopDist, "#.##") +
243         " risk=" + str.toString(stopDist, "#.##"), alert.freq_once_per_bar_close)
244
245 if shortSignal and openShorts < maxOpen and not na(qtyBase) and qtyBase > 0
246     seq += 1
247     string id = "S" + str.toString(seq)
248     strategy.entry(id, strategy.short, qty = qtyBase)
249     _register(id, stopDist, close, 1)
250     if showLabels
251         label.new(bar_index, high, "S", style = label.style_label_down,
252             color = color.new(color.red, 20), textcolor = color.white, size = size.tiny)
253     alert("AURUM_V5 SHORT entry=" + str.toString(close, "#.##") +
254         " stop=" + str.toString(close + stopDist, "#.##") +
255         " risk=" + str.toString(stopDist, "#.##"), alert.freq_once_per_bar_close)
256
257 if sigCT and openShorts < maxOpen and not na(qtyCT) and qtyCT > 0
258     seq += 1
259     string id = "C" + str.toString(seq)
260     strategy.entry(id, strategy.short, qty = qtyCT)
261     _register(id, stopDist, close, 2)
262     if showLabels
263         label.new(bar_index, high, "CT", style = label.style_label_down,
264             color = color.new(color.orange, 20), textcolor = color.white, size = size.tiny)
265     alert("AURUM_V5 CT-SHORT entry=" + str.toString(close, "#.##") +
266         " stop=" + str.toString(close + stopDist, "#.##"), alert.freq_once_per_bar_close)
267

```

```

268 // =====
269 // EXITS - per position, ratcheting on CLOSED BARS ONLY
270 //
271 // The ratchet reads the completed bar's close, never its running high/low. That
272 // is deliberate and is why calc_on_every_tick is false. Trailing from an
273 // intrabar extreme lets the stop jump to a level the trade never actually held;
274 // in testing that produced a +0.089 R phantom edge on coin-flip entries with
275 // costs switched off, where the true answer is mathematically zero.
276 // =====
277 if strategy.opentrades > 0
278     for i = 0 to strategy.opentrades - 1
279         string id = strategy.opentrades.entry_id(i)
280         if map.contains(riskOf, id)
281             float rp = map.get(riskOf, id)
282             float ep = map.get(entryOf, id)
283             int eb = map.get(barOf, id)
284             int kind = map.get(kindOf, id)
285             int dir = kind == 0 ? 1 : -1
286
287             if rp > 0
288                 float excursionR = (close - ep) * dir / rp
289                 float stopLvl = ep - rp * dir // initial stop
290                 float tgtLvl = na
291
292                 if kind == 0 // LONG: full ladder
293                     if excursionR >= r3Trig
294                         stopLvl := ep + r3Lock * rp
295                     else if excursionR >= r2Trig
296                         stopLvl := ep + r2Lock * rp
297                     else if excursionR >= r1Trig
298                         stopLvl := ep + r1Lock * rp
299
300                 else if kind == 1 // SHORT: chosen style
301                     if shortExitMode == "Mirrored ladder"
302                         if excursionR >= r3Trig
303                             stopLvl := ep - r3Lock * rp
304                         else if excursionR >= r2Trig
305                             stopLvl := ep - r2Lock * rp
306                         else if excursionR >= r1Trig
307                             stopLvl := ep - r1Lock * rp
308                     else if shortExitMode == "Tight 1R target"
309                         tgtLvl := ep - rp
310                     else // BE at 1R then 2R target
311                         tgtLvl := ep - 2.0 * rp
312                         if excursionR >= 1.0
313                             stopLvl := ep
314
315                 else // COUNTER-TREND short
316                     tgtLvl := ep - ctTpR * rp
317
318                 if na(tgtLvl)
319                     strategy.exit("x" + id, from_entry = id, stop = stopLvl)
320                 else
321                     strategy.exit("x" + id, from_entry = id, stop = stopLvl, limit = tgtLvl)
322
323                 if bar_index - eb >= maxHoldBars
324                     strategy.close(id, comment = "time stop")
325
326 // =====
327 // DISPLAY
328 // =====
329 bgcolor(tintBg and macroBull ? color.new(color.green, 94) :
330 tintBg and macroBear ? color.new(color.red, 94) : na, title = "Regime")
331
332 plotshape(longSignal, "Long signal", shape.triangleup, location.belowbar,
333 color.new(color.green, 0), size = size.tiny)
334 plotshape(shortSignal, "Short signal", shape.triangledown, location.abovebar,
335 color.new(color.red, 0), size = size.tiny)
336 plotshape(sigCT, "CT signal", shape.xcross, location.abovebar,
337 color.new(color.orange, 0), size = size.tiny)
338
339 var table t = table.new(position.top_right, 2, 8, border_width = 1)
340 if barstate.islast and showPanel
341     table.cell(t, 0, 0, "Aurum Edge V5", text_color = color.white,
342 bgcolor = color.new(color.teal, 25))
343     table.cell(t, 1, 0, "", bgcolor = color.new(color.teal, 25))
344     table.cell(t, 0, 1, "Daily regime")
345     table.cell(t, 1, 1, macroBull ? "BULL" : macroBear ? "BEAR" : "neutral",
346 text_color = macroBull ? color.green : macroBear ? color.red : color.gray)
347     table.cell(t, 0, 2, "Long engine")
348     table.cell(t, 1, 2, enableLong ? (macroBull ? "armed" : "idle") : "off")
349     table.cell(t, 0, 3, "Short engine")
350     table.cell(t, 1, 3, enableShort ? (macroBear ? "armed" : "idle") : "off (tested -0.086R)")
351     table.cell(t, 0, 4, "Counter-trend")
352     table.cell(t, 1, 4, enableCT ? "armed" : "off (tested -0.21R)")
353     table.cell(t, 0, 5, "Open L / S")
354     table.cell(t, 1, 5, str.tostring(openLongs) + " / " + str.tostring(openShorts))
355     table.cell(t, 0, 6, "Stop ($) / cost%")
356     table.cell(t, 1, 6, str.tostring(stopDist, "#.##") + " / " +

```

```
357         str.toString(0.45 / stopDist * 100, "#.#") + "%")
358     table.cell(t, 0, 7, "Closed trades")
359     table.cell(t, 1, 7, str.toString(strategy.closedtrades))
360
361 // =====
362 // ALERT CONDITIONS (for automation wiring)
363 // =====
364 alertcondition(longSignal, "Aurum V5 Long entry", "Aurum Edge V5: LONG {{ticker}} @ {{close}}")
365 alertcondition(shortSignal, "Aurum V5 Short entry", "Aurum Edge V5: SHORT {{ticker}} @ {{close}}")
366 alertcondition(sigCT, "Aurum V5 CT short", "Aurum Edge V5: COUNTER-TREND SHORT {{ticker}} @ {{close}}")
```